Lab 7 — Fuzzing for Input Validation Bugs

Laundry Service • SETAPESU26/L6_L7_41_laundry_service • Hypothesis

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Step 2 — Run the buggy application

The program works for well-formed input. The second run feeds it **regular** with no colon and it dies with an unhandled `ValueError` from `text.split(':')` — a user typo takes the whole program down.

```
PS C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code> python laundry_service.py
==== Running parse_order ====
Enter garment:weight in kg (e.g. regular:6): Garment and Weight: ('Regular', 6.0)

==== Running weight_average ====
Enter garment weights as CSV (e.g. 3,5,8): Average Weight: 5

==== Running payable_charge ====
Enter charge|discount (e.g. 300|40): Payable Charge: 314.0

PS C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code> python laundry_service.py   # same program, malforme
    d input
==== Running parse_order ====
Enter garment:weight in kg (e.g. regular:6): Traceback (most recent call last):
  File "C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code\laundry_service.py", line 31, in <module>
    main()
    ~~~~^^
  File "C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code\laundry_service.py", line 17, in main
    result = parse_order(user_input)
  File "C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code\laundry_service.py", line 2, in parse_order
    garment, kg = text.split(':')
    ^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Step 4 — Run the Hypothesis suite against the buggy code

5 failed, 4 passed. Hypothesis minimised each failure to the smallest input that triggers it: `''` and `None` for every function, plus `'0'` finding the missing separator. These are inputs a hand-written test suite would not have thought to try.

```
PS C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code> pytest test_laundry_service.py
    |   File "C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code\laundry_service.py", line 10, in payable_c
    harge
    |     charge, discount = text.split('|')
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_payable_charge_is_never_negative_or_nan(
    |     s='',
    | )
    +---------------- 4 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code\test_laundry_service.py", line 112, in tes
    t_payable_charge_is_never_negative_or_nan
    |     result = payable_charge(s)
    |   File "C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code\laundry_service.py", line 10, in payable_c
    harge
    |     charge, discount = text.split('|')
    |                        ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_payable_charge_is_never_negative_or_nan(
    |     s=None,
    | )
    +------------------------------------
========================== short test summary info ==========================
FAILED test_laundry_service.py::test_parse_order_no_crash - ExceptionGroup: H...
FAILED test_laundry_service.py::test_parse_order_shape - ExceptionGroup: Hypo...
FAILED test_laundry_service.py::test_weight_average_no_crash - ExceptionGroup...
FAILED test_laundry_service.py::test_payable_charge_no_crash - ExceptionGroup...
FAILED test_laundry_service.py::test_payable_charge_is_never_negative_or_nan
5 failed, 4 passed in 2.62s
```

Step 5 — Run the hardened application

Identical output for valid input — the fix changed no behaviour that worked before. The same malformed input that crashed in Step 2 is now reported and skipped.

```
PS C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code> python laundry_service.py
==== Running parse_order ====
Enter garment:weight in kg (e.g. regular:6): Garment and Weight: ('Regular', 6.0)

==== Running weight_average ====
Enter garment weights as CSV (e.g. 3,5,8): Average Weight: 5

==== Running payable_charge ====
Enter charge|discount (e.g. 300|40): Payable Charge: 314.0

PS C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code> python laundry_service.py   # same malformed input a
    s before
==== Running parse_order ====
Enter garment:weight in kg (e.g. regular:6): Could not read that as garment:weight - skipping.

==== Running weight_average ====
Enter garment weights as CSV (e.g. 3,5,8): Could not read that as a list of whole weights - skipping.

==== Running payable_charge ====
Enter charge|discount (e.g. 300|40): Could not read that as charge|discount - skipping.
```

Step 6 — Re-run the Hypothesis suite against the fixed code

9 passed. Hypothesis generated hundreds of inputs per property — text, None, integers, floats, booleans, lists and raw bytes — and none of them produced an exception.

```
PS C:\Users\prajw\Desktop\SE-Labs-PES1UG24CS590\Lab-7\code> pytest test_laundry_service.py
.........                                                                [100%]
9 passed in 2.27s
```